

Basic Principles of Medicine 2

Module: Digestion and Metabolism

Lecture DM 34

The Feed-Fast Cycle: Role of Hormones

Dr. Shellon Thomas
sthoma10@sgu.edu

Acknowledgment: Dr. Margit Trotz, Ph.D.

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Recommended Reading

- Lippincott Illustrated Reviews Biochemistry (**Lippincott**) 9th edition; Chapter 24: Pages 357 – 372; [The Feed–Fast Cycle | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Lippincott's Biochemistry 9th Edition
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lww.healthlibrary.com/book.aspx?bookid=3402>

Objectives: The Feed-Fast Cycle: Role of Hormones

SOM. MK.III.BPM2.3.DM.3.BCHM.1320	Describe roles of insulin and glucagon and insulin:glucagon ratio in regulation of carbohydrate, lipid and protein metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1321	Describe roles of epinephrine and glucocorticoid in regulation of fuel metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1322	Differentiate between postprandial phase, post-absorptive phase and early phase of starvation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1323	Describe pathways leading to fatty synthesis and cholesterol synthesis in the fed state starting with glucose.
SOM. MK.III.BPM2.3.DM.3.BCHM.1324	Describe effect of changes in plasma epinephrine levels on carbohydrate, lipid and protein metabolism.
SOM. MK.III.BPM2.3.DM.3.BCHM.1325	Outline metabolic changes that occur during the feed-fast cycle.
SOM. MK.III.BPM2.3.DM.3.BCHM.1326	Describe changes in plasma glucose levels in the feed-fast cycle and during prolonged starvation.
SOM. MK.III.BPM2.3.DM.3.BCHM.1327	Describe regulation and coordination of metabolic pathways during the feed-fast cycle in the various tissues.

Roles of Insulin and glucagon in homeostasis

- They regulate fuel storage and mobilization
- Their blood levels are continuously fluctuating
- Their plasma half-lives are in the range of minutes
- Both released and stimulated by arginine to assure that both hormones are present in blood
- Serum insulin: glucagon ratio determines metabolic status

Pathways activated by Insulin and glucagon

Insulin activates:

- Glycolysis
- Glycogen synthesis
- Protein synthesis
- Synthesis of fatty acids
- Synthesis of cholesterol

Glucagon activates in the liver:

- Glycogen degradation
- Gluconeogenesis (uses amino acids)
- Oxidation of fatty acids
- Synthesis of ketone bodies

Metabolic functions of insulin and glucagon

Insulin (major anabolic hormone) → promotes storage of fuel and growth

- Glycogenesis in liver and muscle
- Fatty acid and TAG synthesis in liver (release of VLDL)
- Protein synthesis in muscle and liver
- TAG synthesis in fat cells

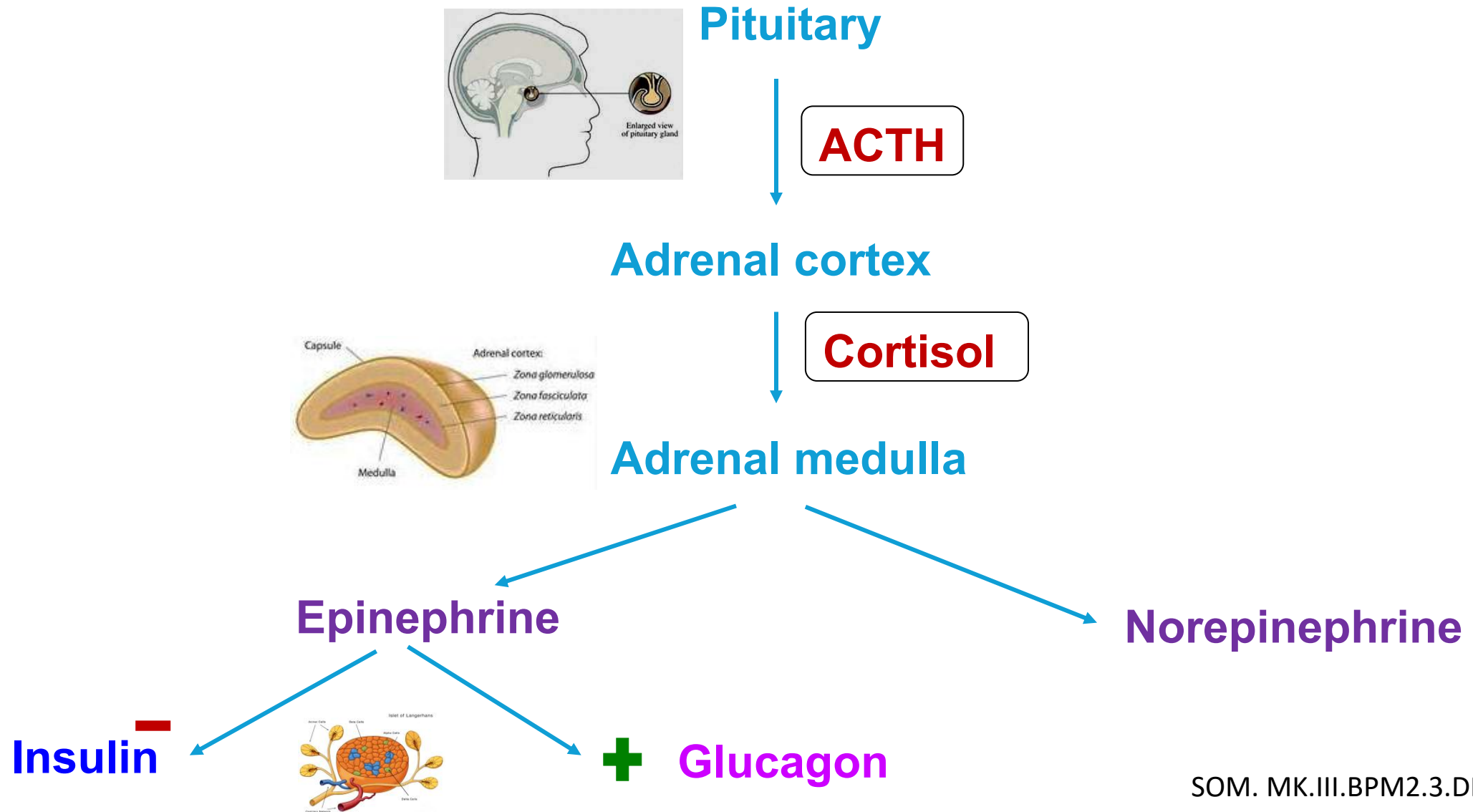
Glucagon → fuel mobilization

- Hepatic glycogenolysis and gluconeogenesis
- Hepatic ketone body synthesis and release
- Renal gluconeogenesis
- TAG degradation in fat cells (lipolysis) → release of free fatty acids and glycerol into blood at low insulin: glucagon ratio

Glucose role in metabolic homeostasis

- Many cells (especially BRAIN) depend on glucose (glycolysis) for energy
- Blood glucose level is tightly controlled
- Insulin and glucagon: Hormonal regulators of blood glucose levels
- Release responds directly to blood glucose level
- Epinephrine, cortisol, growth hormone: Insulin counterregulatory hormones
 - Released to prevent hypoglycemia

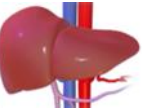
Stress situations overrule other regulations



Roles of cortisol and epinephrine in fuel metabolism

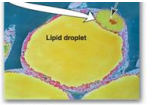
1. Cortisol

- **Skeletal muscle protein degradation** → release of amino acids into blood (for hepatic gluconeogenesis)
- Acts with glucagon for **long-term hepatic gluconeogenesis** → **hyperglycemia**



2. Epinephrine

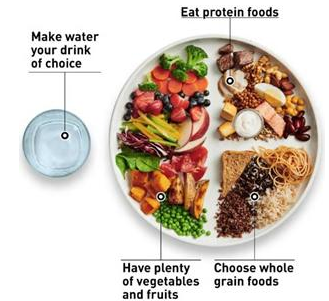
- **TAG degradation in fat cells (Lipolysis)** → release of free fatty acids (used by cells for energy) + glycerol (for hepatic gluconeogenesis)
- Activates **hepatic glycogen degradation** (with glucagon) → Hyperglycemia



General time frame of the fed and fast state

- **Postprandial (absorptive) phase**

- Fed state
- Digestion and absorption after a meal (prandi)
- Depends on diet (glycemic index and amount of carbohydrate)
- **2 hours after food intake**
- Normal post-prandial blood glucose: less than 140 mg/dL after 2 hours



- **Fasting** (Post-absorptive phase)

- Fasting state (Usually 8-10 hours after last meal)
- Food is completely digested and absorbed (Approximately 5 hours after a meal)
- Normal fasting blood glucose: 70-100 mg/dL (5mM/L)

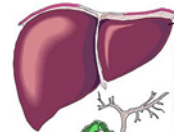


- **Early phase of starvation**

- 24 hours - **3 days** after food intake
- Plasma glucose: Further **decline** to 70 mg/dL (about 3.8 mM)
- Plasma glucose maintained even **after 40 days** of starvation

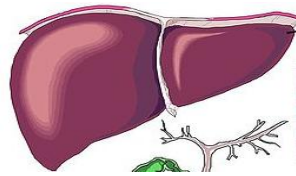
- **Prolonged starvation** → greater than 10 days of fasting/starvation

Liver metabolism in Post-prandial phase



- Liver receives blood from portal vein containing the dietary monosaccharides
- Glucokinase (high K_m) facilitates glucose trapping in hepatocytes
- $\uparrow\uparrow$ blood glucose levels $\rightarrow$ insulin secretion
- Insulin favors
 - Glycolysis and glycogenesis in liver
 - Stimulates use of glucose for fatty acid and cholesterol synthesis in liver
 - VLDLs are filled with TAG and cholesteryl esters $\rightarrow$ released into blood
- Glucose uptake into **fat and muscle cells** using GLUT-4
- Insulin actions on liver, adipose tissue, and muscle $\rightarrow$ Eventual lowering of blood glucose level

After a meal



Glucose

Glucokinase

Glucose 6-P

Glycolysis

NADPH used for FA
and cholesterol
synthesis

PPP

**Glycogen
synthesis**

PDH

TCA cycle

NADH
FADH₂

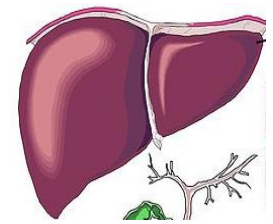
Citrate in cytosol

**Fatty acid synthesis
Cholesterol synthesis
TAG synthesis
and formation of VLDL**

**Release
of VLDL**

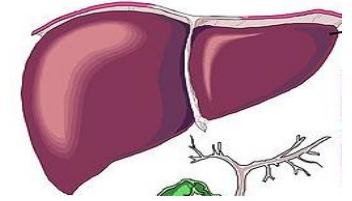
BLOOD
**Uptake of
Glucose**

Blood glucose homeostasis in the feed-fast cycle: Functions of the liver



- “Buffers” blood **glucose** levels
- **Glucose** flows with the gradient in **both directions** via **GLUT-2**
- After a meal: **hepatocytes reduce** elevated blood **glucose levels**

Postprandial Metabolism: Effect of high insulin levels on the liver



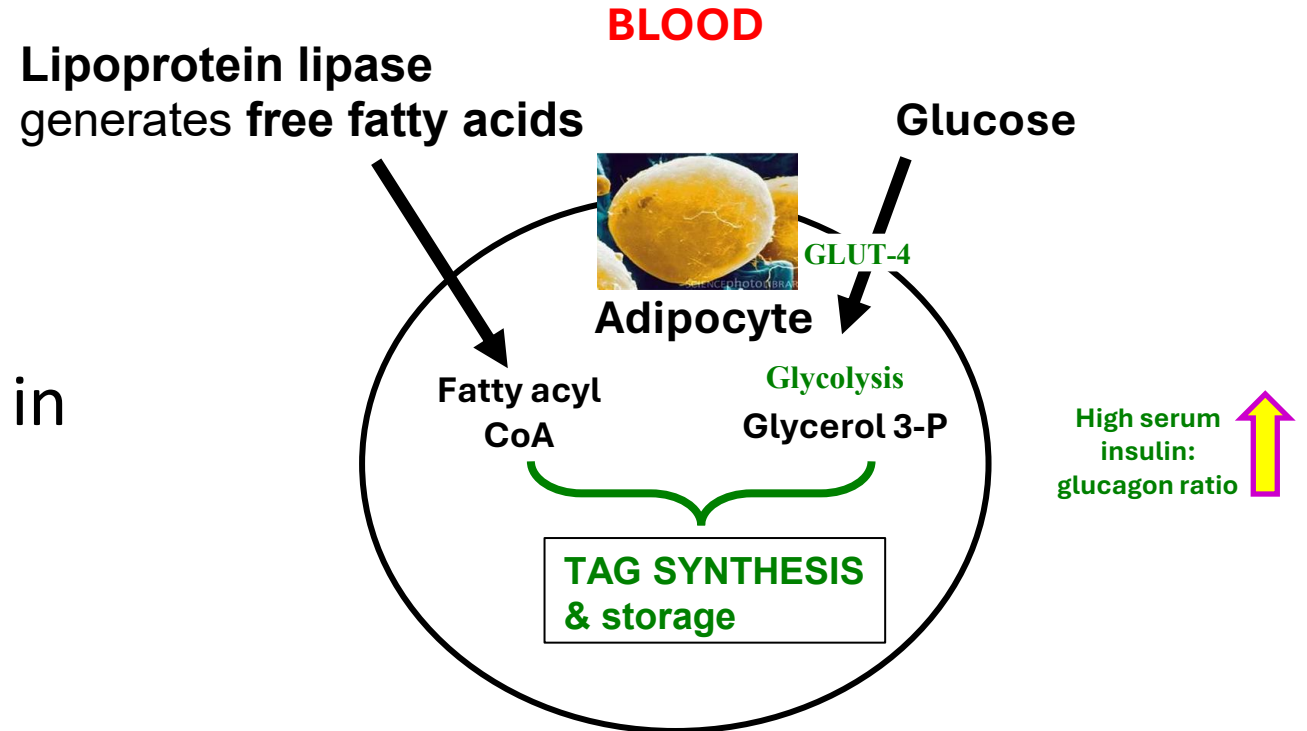
- **Glycogenesis is active** → **Glycogenolysis is inhibited**
- **Glycolysis is active** → **Gluconeogenesis is inhibited**
- **Fatty acid synthesis is active** → **β -oxidation is reduced**
- **PPP is active.**

(NADPH is used in the synthesis of fatty acids and of cholesterol).

- **PDH is active** and connects glycolysis to the TCA cycle
- **TCA cycle is active**
- **Cholesterol** synthesis and **TAG** synthesis are active and **VLDLs** are released into the **blood**.

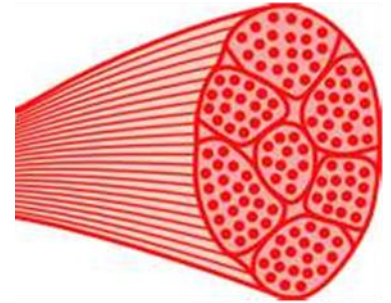
Postprandial adipose tissue metabolism: High insulin

- After a meal insulin:
 1. Activates **lipoprotein lipase** in capillaries of adipose tissue
 2. Increases GLUT-4 transporters in the fat cell plasma membrane
 3. Stimulates TAG synthesis in fat cells
 4. **Hormone-sensitive lipase less active**
- Glucose is needed for the synthesis of the glycerol **backbone** of **TAGs**



Postprandial Muscle Metabolism

High insulin levels



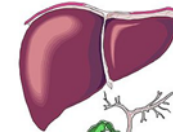
1. Activates glucose uptake from the blood (via GLUT-4)
→ Glycogen synthesis and glycolysis take place
2. Activates amino acid uptake from the blood
→ Protein synthesis is stimulated especially by leucine

Note: **Branched-chain amino acids** used for protein synthesis and energy metabolism

Metabolism in Post-absorptive (fasting) phase

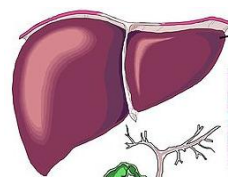
- Blood glucose levels lower than in post-prandial state due to metabolic actions of insulin
- ↓ blood glucose → **release of counterregulatory hormones** and ↓ insulin release (**low insulin: glucagon ratio**)
- The following is released into the blood:
 1. **Glucose** (hepatic glycogenolysis and hepatic and renal gluconeogenesis)
 2. **Free fatty acids and free glycerol** (TAG degradation/lipolysis in fat cells favored by low insulin: glucagon ratio and epinephrine)
 3. **Amino acids from skeletal muscle degradation** (low insulin: glucagon ratio and cortisol)

Liver metabolism during fasting

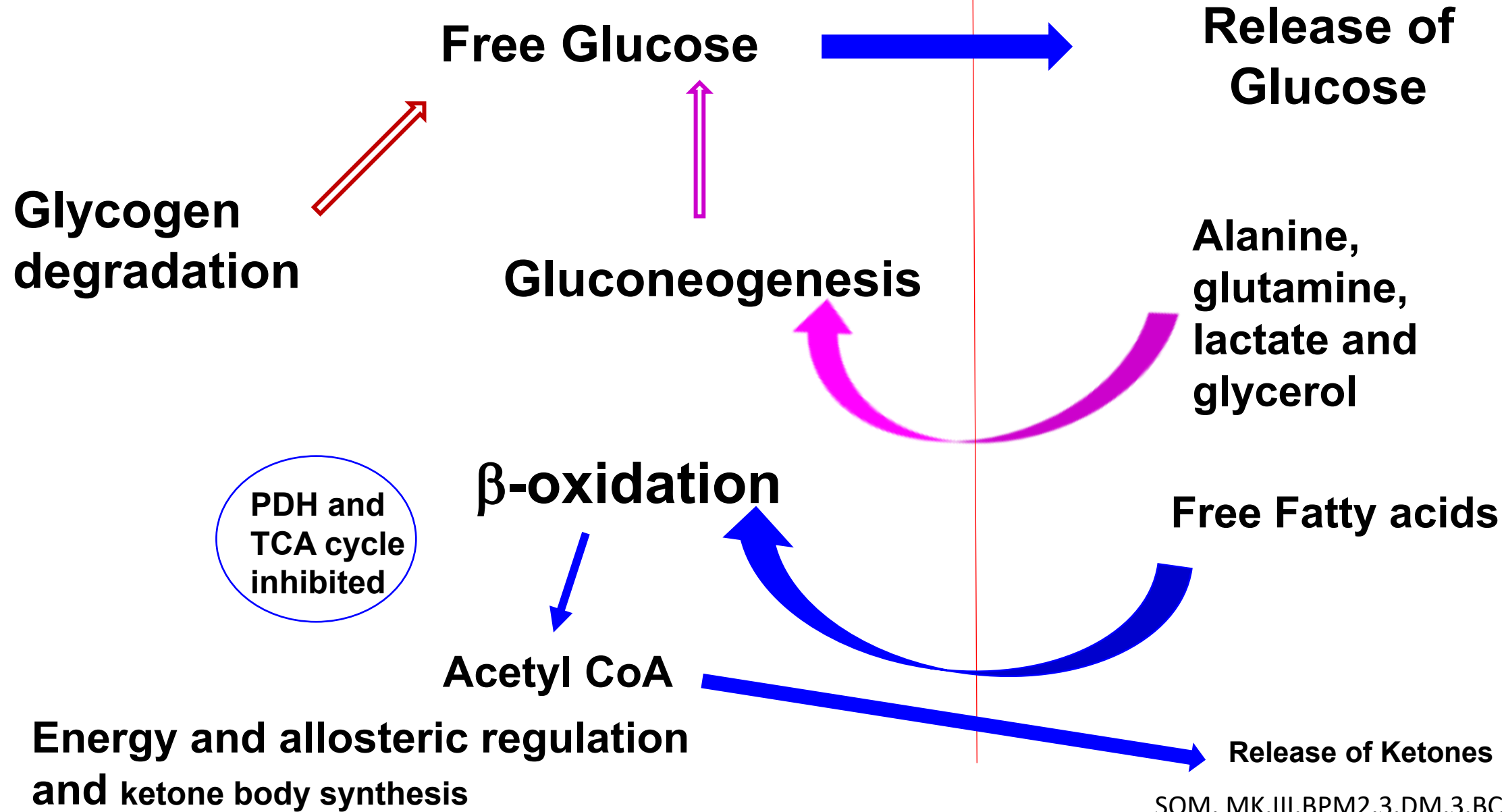


- **Pathways activated:**
- **Low insulin: glucagon ratio** favors **Glycogen degradation** and **gluconeogenesis**
 - ✓ Glycogen degradation (glucagon, epinephrine)
 - ✓ Gluconeogenesis (glucagon and cortisol; Uses amino acids from muscle proteolysis)
 - ✓ Glucose released into the blood
 - ✓ Fatty acid beta-oxidation (CPT-1 active)
 - ✓ Ketogenesis
- **Pathways inhibited**
 - Glycogen synthesis
 - Glycolysis
 - Synthesis of fatty acids and cholesterol

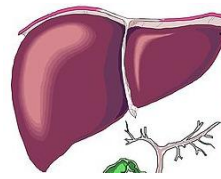
Overnight Fast



BLOOD



Starvation (>1 day)



BLOOD

Glycogen
depleted

Free Glucose

**Release of
Glucose**

Gluconeogenesis

Alanine,
glutamine,
lactate and
glycerol

PDH and
TCA cycle
inhibited

β -oxidation

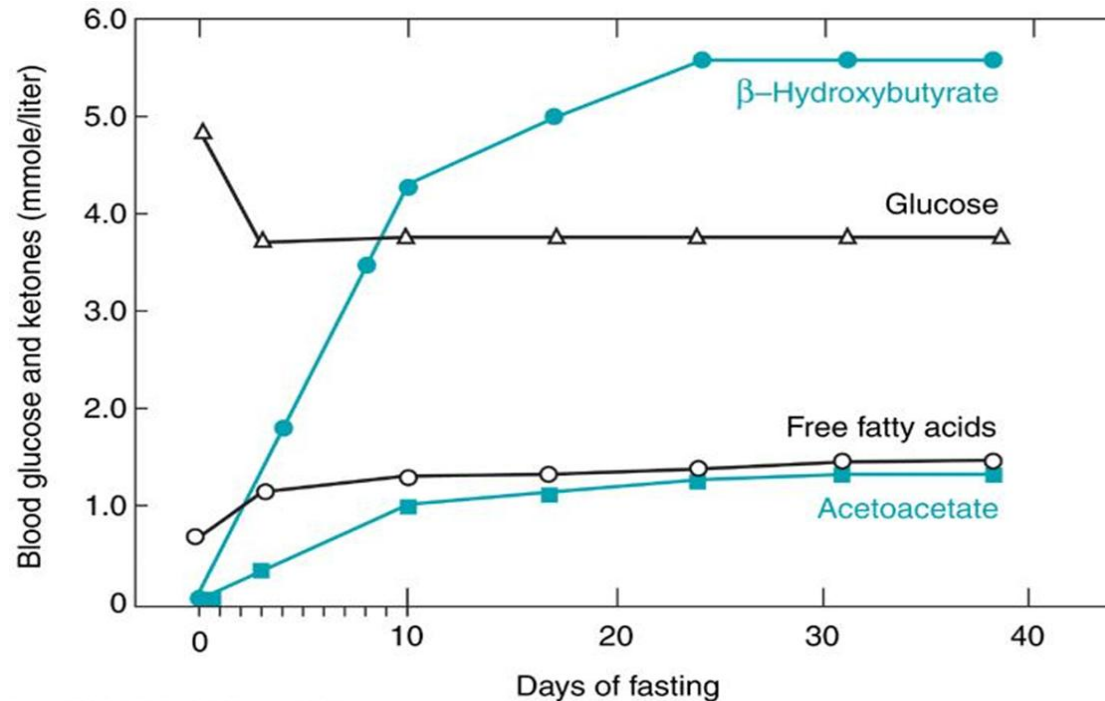
Fatty acids

Acetyl CoA

**Release of
Ketones**

**Energy and allosteric regulation
and ketone body synthesis**

Changes in metabolite levels in fasting



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Glucose: Reduced to 3.8 mM at day 3: Due to ketone body utilization by brain in prolonged fasting

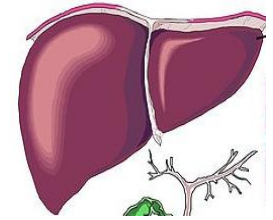
Fatty acids: Increased and plateau after day 3: Due to Hormone-sensitive lipase (Lipolysis)

Acetoacetate: Increased: Due to adipose tissue lipolysis and hepatic ketogenesis

Hydroxybutyrate: Highly elevated (Major ketone body): Due to adipose tissue lipolysis and ketogenesis

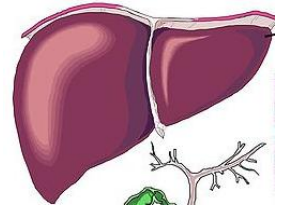
Blood glucose homeostasis in fasting:

Functions of liver



- During **fasting**:
- **Hepatocytes** release **glucose** into blood (via GLUT 2)
- In **liver** and renal cortex, **NADH** and **acetyl CoA**: **Allosteric activators** of **gluconeogenesis**
- **Ketone bodies** have a “protein-sparing effect” in **starvation**
 - Less gluconeogenesis from amino acids is needed as ketone bodies are used instead of glucose by the brain
 - Hence, use of ketone bodies by the brain reduces requirement of amino acids for gluconeogenesis

Fasting: Effect of high glucagon level on the liver



Glycogen degradation is active

Gluconeogenesis is active

β -oxidation is active

Ketone body synthesis is active

Glycogen synthesis is inhibited

Glycolysis is inhibited

PPP is reduced

PDH is inhibited

TCA cycle is inhibited

Fatty acid synthesis is inhibited

Cholesterol synthesis is inhibited

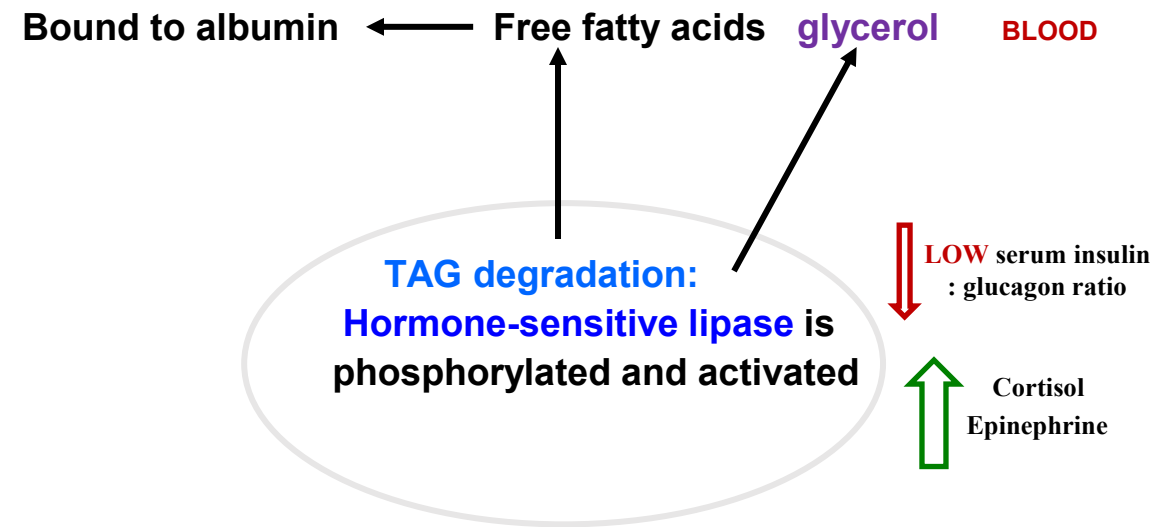
TAG synthesis is inhibited



Fasting: Adipose tissue metabolism



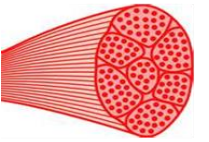
- TAGs degraded in fat cells
→ **Hormone-sensitive lipase** active → Release of free fatty acids and glycerol into blood
- **Glycerol**: used for **gluconeogenesis** in liver
- **Fatty acids**: used for **energy metabolism** (muscle and liver) and ketogenesis (liver)





Fasting: Muscle Metabolism: ↓ **insulin** and ↑ **cortisol**

- **Uptake of free fatty acids and ketone body** from blood → energy metabolism in oxygenated, resting muscle
- **Protein degradation** stimulated by **low insulin** and increased cortisol.
→ Amino acids (mainly alanine and glutamine) released into blood → Used for **gluconeogenesis** to maintain plasma glucose at ~3.8 mM
- Activates **alanine-glucose cycle**



Brain metabolism: Fasting



- **Glucose:** always needed for energy and **synthesis of neurotransmitters**
- **Ketone bodies** (>3 days starvation) → used in addition to aerobic glycolysis for **energy metabolism**
- Uptake of glucose is mostly insulin-independent and performed mainly by GLUT-1
- Most fatty acids cannot pass the adult blood-brain barrier
 - except **dietary essential fatty acids** (linoleic and linolenic acid)(In the developing brain, arachidonic acid and DHA can also pass)

Summary

- Hormones and availability of substrates determines pathways in liver, muscle, and fat cells

- **After a meal:** ↑↑ **blood glucose**

- ✓ **Liver:** Glycogenesis, glycolysis, PPP, fatty acid, cholesterol and TAG synthesis, and VLDL formation
- ✓ **Muscle:** Glycolysis and glycogen synthesis
- ✓ **Fat cells:** Glycolysis and TAG synthesis
- ✓ **Brain:** Glycolysis



- **During fasting:** ↑↑ **blood levels of free fatty acids and ketone bodies**

- ✓ **Liver:** β -oxidation of fatty acids and ketone body synthesis
- ✓ **Muscle:** β -oxidation of fatty acids and ketone body use
- ✓ **Brain:** Glucose and Ketone body use



REVIEW



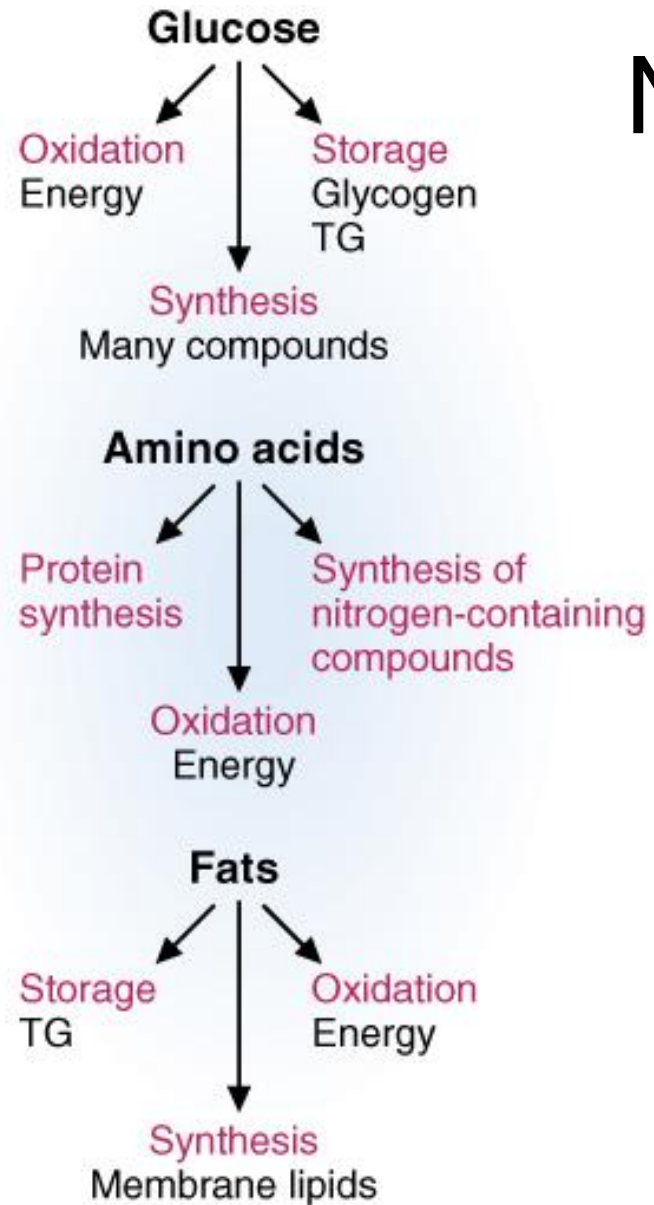
Insulin



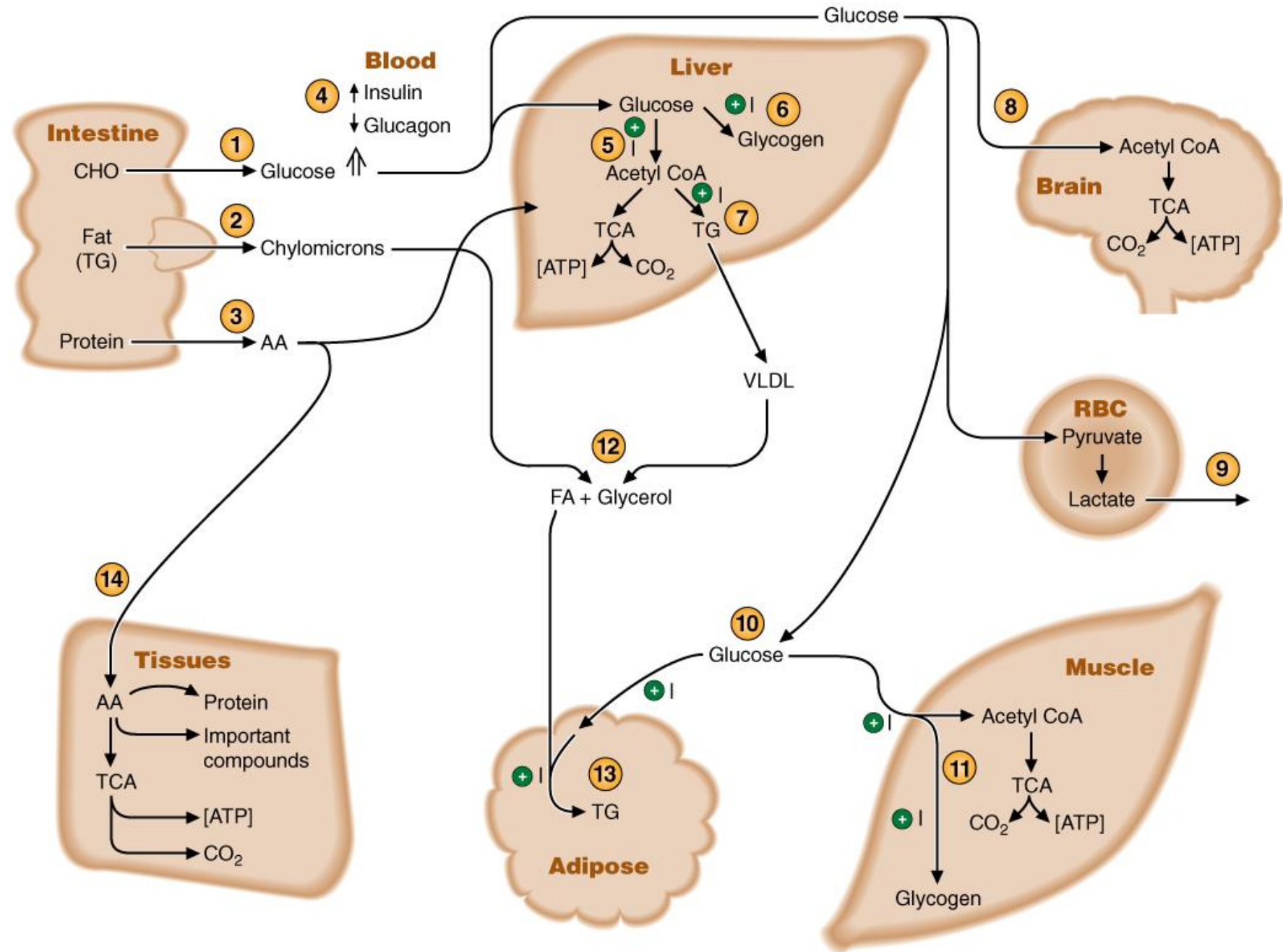
Glucagon

The following slides are **not** part of this lecture but they
provide a more detailed review and connection to other
lectures.

Major fates of fuel in fed state



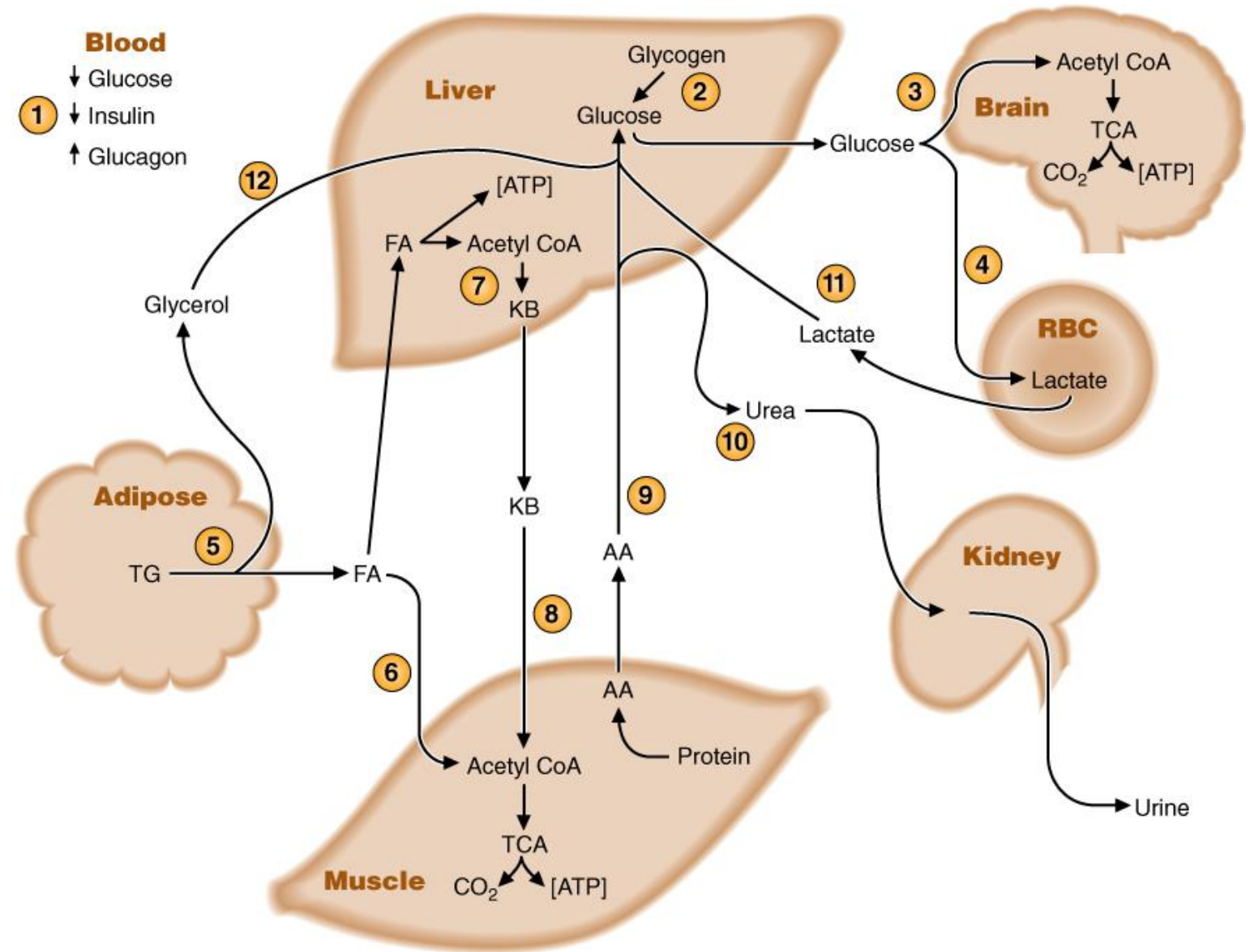
Metabolism in Fed state



Metabolism in Fed state (refer to numbers on previous slide)

1. Absorption of dietary glucose increases plasma glucose
2. Dietary TAG are absorbed and transported as chylomicrons
3. Dietary protein is absorbed as amino acids
4. Increased plasma glucose results in insulin secretion and decreases glucagon secretion (Increases insulin: glucagon ratio)
5. Liver glycolysis is stimulated
6. Liver glycogenesis is favored
7. Fatty acid synthesis from acetyl CoA is stimulated and triacylglycerol (TAG) formation is stimulated
8. Brain uses plasma glucose as an energy source (complete oxidation to CO₂ and H₂O)
9. RBCs uses glucose as an energy source to form lactate which is transported to liver by Cori cycle
10. Glucose entry into adipocytes is favored due to presence of GLUT-4 on adipocyte membrane
11. Glucose entry into muscle is favored (GLUT-4) and glycogen synthesis in muscle is stimulated
12. Lipoprotein lipase present in endothelium of blood vessels of adipocytes and muscle is active and forms Fatty acids and glycerol which are taken up by adipocytes.
13. In adipocytes, formation and storage of TAG is favored. Glycerol-phosphate is provided by DHAP from glycolysis.
14. Amino acid entry and uptake into all tissues is favored. Protein synthesis is active

Metabolism 10-12 hours after meal (Overnight fast)



Metabolism 10-12 hours after meal (Overnight fast) (refer to numbers on previous slide)

1. A fall in plasma glucose reduces insulin: glucagon ratio
2. Liver glycogenolysis is stimulated
3. Plasma glucose is used for brain metabolism. Glucose is completely oxidized to CO₂ and H₂O
4. RBCs use glucose and form lactate.
5. Adipose tissue lipolysis is active (TAG broken down to Free fatty acids and glycerol by hormone sensitive lipase)
6. Fatty acids are used for energy by skeletal and cardiac muscle
7. Fatty acid oxidation in the liver provides energy to the liver and forms ketone bodies
8. Ketone bodies are used as sources of energy by the muscle
9. Muscle Proteolysis is active and releases amino acids. The amino acids in liver are used for gluconeogenesis.
10. Ammonia generated by amino acid metabolism is detoxified to Urea which is lost through the kidneys (urine)
11. Cori cycle transports lactate to liver which is used for gluconeogenesis in liver.
12. Glycerol formed in adipose tissue is also used as a precursor of gluconeogenesis

Postprandial Liver Metabolism (1): Carbohydrate metabolism

Glycogen synthesis is active	Glycogen synthase is dephosphorylated and active.
Glycogen degradation is inhibited.	Glycogen phosphorylase kinase and glycogen phosphorylase are dephosphorylated and inactive.
PPP is active.	Glucose 6-P DH is active and forms NADPH, which is used to synthesize fatty acids and cholesterol.
Glycolysis is active.	Bifunctional enzyme is dephosphorylated: active PFK-2 forms fructose 2,6-bisP which overcomes ATP-inhibition of PFK-1. Pyruvate kinase is dephosphorylated and active
Gluconeogenesis is inhibited.	Fructose1,6-bisphosphatase is inhibited by fructose 2,6-bisP.

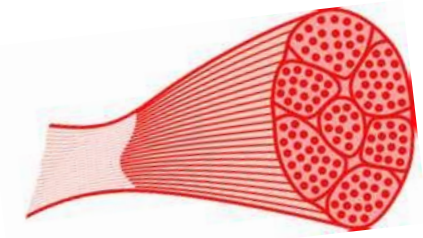
Postprandial Liver Metabolism (2): Lipid metabolism

Hepatocytes perform synthesis of **fatty acids**, **TAGs** and **cholesterol** at high blood glucose levels.

PDH is active	Acetyl CoA used in TCA cycle and citrate formation; NADH → ETC
TCA cycle is active	NADH and FADH ₂ → ETC. Citrate used for fatty acid and cholesterol synthesis
β-oxidation is inhibited	CPT-I of the carnitine shuttle is inhibited by cytosolic malonyl CoA
Fatty acid synthesis is active	Acetyl CoA carboxylase is dephosphorylated and active. Cytosolic citrate leads to acetyl CoA carboxylase polymerization and activation.
Cholesterol synthesis is active.	HMG CoA reductase is dephosphorylated and active.
TAG synthesis is active	From <i>denovo</i> fatty acids synthesis
Synthesis and release of VLDLs into the blood	

Postprandial resting skeletal muscle metabolism

Review



Insulin

Insulin activates:

- **GLUT-4** alignment in cell membrane → **glucose uptake** → ↓ blood **glucose** levels.
- **Glucose** is used for muscle **glycogen synthesis** and **glycolysis**.
- **Stimulates uptake of amino acids** → **protein synthesis**
- Branched-chain amino acids also used for energy metabolism in muscle

Fasting Liver Metabolism (1): Carbohydrate metabolism

Liver: Release of glucose into the blood by both glycogen degradation and gluconeogenesis

Glycogen synthesis is inhibited.	Glycogen synthase is phosphorylated by PKA and inactive.
Glycogen degradation is activated by glucagon and epinephrine.	Glycogen phosphorylase kinase is phosphorylated by the activated PKA. Glycogen phosphorylase kinase phosphorylates and activates glycogen phosphorylase.
Glycolysis is inhibited.	Bifunctional enzyme is phosphorylated by PKA: active Bpase-2 degrades fructose 2,6-bisP. PFK-1 is inhibited by normal ATP level. Hepatic pyruvate kinase is phosphorylated by PKA and inhibited. Hepatic pyruvate kinase is also allosterically inhibited by alanine and ATP.
Gluconeogenesis is active.	Fructose 1,6-bisphosphatase is not inhibited by fructose 2,6-bisP. Pyruvate carboxylase is activated by acetyl CoA from β -oxidation.
PDH is inhibited	by acetyl CoA, NADH and PDH kinase which phosphorylates PDH.
TCA cycle (isocitrate DH) is inhibited by NADH from beta-oxidation.	Oxaloacetate is reduced to malate and used for gluconeogenesis

Fasting Liver Metabolism (2): Liver metabolism

Carnitine shuttle active	CPT-I is not inhibited by malonyl CoA and free fatty acids released from fat cells are taken up from the blood and enter mitochondria of hepatocytes for beta-oxidation (fatty acid oxidation)
Fatty acid oxidation (Beta-oxidation)	Active and is regulated by NADH and acetyl CoA levels.
Ketone body synthesis active in hepatic mitochondria:	Mitochondrial HMG CoA synthase and HMG CoA lyase have substrates due to high acetyl CoA. 3-Hydroxybutyrate formation from acetoacetate is stimulated by high NADH level. Acetoacetate decarboxylation can generate acetone (spontaneous). Only Hepatocytes can perform ketone body synthesis (contain mitochondrial HMG CoA synthase). Acetoacetate and 3-hydroxybutyrate are released into blood → Ketone body levels Elevated Liver cannot use ketone bodies as it lacks thiophorase
Fatty acid synthesis is inhibited.	Acetyl CoA carboxylase is phosphorylated by AMP kinase and is inactive. Glucagon activates AMP-kinase
Cholesterol synthesis is inhibited.	HMG CoA reductase is phosphorylated by AMP kinase and is inactive. Glucagon activates AMP-kinase

Glucose transporters

Secondary active transport, sodium-cotransport which needs active sodium/potassium ATPase	
SGLT-1	Small intestine uptake of dietary glucose or galactose .
SGLT-2	Kidney: Reabsorption of glucose into blood from the renal tubules
Facilitated transport using GLUT isoforms	
GLUT-1	High affinity. RBC and blood brain-barrier: basal uptake of glucose. Found in many cells.
GLUT-2	High-capacity, low affinity. Small intestine: release of all dietary monosaccharides into the hepatic portal vein. Liver: uptake of all monosaccharides after a meal from the hepatic portal vein. Liver and kidney: release of glucose during fasting into the blood Beta-cells of pancreas: uptake of glucose for glucose sensing and release of insulin.
GLUT-3	High affinity. Brain, neurons in central nervous system. Found in many cells.
GLUT-4	High affinity, insulin-sensitive . Muscle and adipose tissue: insulin leads to an increase of GLUT-4 in the plasma membrane
GLUT-5	High affinity for fructose. Transport of glucose at very high glucose concentration Small intestine: Absorption of dietary fructose. Testes: release of fructose into semen.

Summary

	Insulin	Glucagon (liver)	Epinephrine
Carbohydrate metabolism	Glycogen synthesis and glycolysis occur in liver and muscle. GLUT-4 for uptake of glucose.	Glycogen degradation and gluconeogenesis occur in liver. (note: gluconeogenesis also in kidney cortex)	Glycogen degradation and glycolysis in muscle. Glycogen degradation in the liver.
Lipid metabolism	Synthesis of fatty acids, TAGs, and cholesterol in liver. Release of VLDL. Adipocyte endothelial Lipoprotein lipase is active and free fatty acids are taken up for TAG synthesis and storage	Hepatic beta-oxidation in mitochondria and active carnitine shuttle. Liver only: Ketone body synthesis and release into blood. Mitochondrial HMG CoA synthase only in liver	Fatty acid oxidation and extrahepatic use of ketone bodies. Lipolysis in fat cells and release of free fatty acids and glycerol into blood. (cortisol has long-term action)
Protein metabolism	Uptake of amino acids and synthesis of proteins in muscle. Induction (synthesis) of key glycolytic enzymes in liver	Use of amino acids for gluconeogenesis. Alanine and glutamine from the blood. Activated alanine-glucose cycle.	Degradation of muscle proteins. Release of alanine and glutamine into the blood. (Mainly after cortisol action)

Required post-reading and objectives

- DLA_Obesity

SOM. MK.III.BPM2.3.DM.3.BCHM.1328	Define obesity using body mass index (BMI)
SOM. MK.III.BPM2.3.DM.3.BCHM.1329	Compare the various methods available for diagnosis of obesity including BMI, waist circumference, Waist to hip ratio, bioelectric impedance and discuss the significance of location of body fat ('apple' vs 'pear') in obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1330	Summarize factors contributing (multifactorial) to obesity: including genetic, environmental, behavioral factors and the role of leptin.
SOM. MK.III.BPM2.3.DM.3.BCHM.1331	Analyze metabolic changes in obesity: including Dyslipidemia (Correlate changes in lipid profile to metabolic changes in obesity) and Syndrome X or Metabolic syndrome (Correlate clinical and biochemical features and explain significance of insulin resistance).
SOM. MK.III.BPM2.3.DM.3.BCHM.1332	Explain nutritional and lifestyle changes advised for the management of obesity.
SOM. MK.III.BPM2.3.DM.3.BCHM.1333	Evaluate the role of surgery and drugs (sibutramine and orlistat) in obesity management.

**Thank you, if you have questions,
please email**

Dr. Shellon Thomas
sthoma10@sgu.edu